CUT&Tag Method

1. Sample collection and preparation(for Cell samples)

Take out the cryovial and shake it in 37 °C water to quickly thaw it within 1-2 min.Cells were centrifuged for 5 min at 500×g at 4 ℃ Cell activity was detected with LUNA-FL and counted.

2. Sample collection and preparation(for Tissue samples)

1 ml of 1 XHB buffer was added into the tissue sample, grinding and filtering, filtrate centrifuged at 4 ℃ 500×g for 5 min, supernatant was discarded, density gradient centrifugation for 10 min, nucleus suspension was obtained and counted with LUNA-FL.

3. Library construction for CUT&Tag

CUT&Tag library was performed as described previously [1]. Briefly, the cells are bound to Concanavalin A-coated magnetic beads, and the cell membrane is permeabilized by Digitonin. The enzyme pA-Tn5 Transposase precisely binds the DNA sequence near the target protein under the antibody guidance and results in factor-targeted tagmentation. DNA sequence is tagmented, with adapters added at the same time at both ends, which can be enriched by PCR to form the sequencing-ready libraries(CUT&Tag_Novogene Kit). After the PCR reaction, libraries were purified with the AMPurebeads and library quality was assessed on the Agilent Bioanalyzer 2100 system.

4. Sequencing

The clustering of the index-coded samples was performed on a cBot Cluster Generation System using TruSeq PE Cluster Kit v3-cBot-HS (Illumina)according to the manufacturer’ s instructions. The library preparations weresequenced on Illumina Novaseq platform at Tianjin Novogene Bioinformatic Technology Co., Ltd.(Beijing, China)and 150 bp paired-end reads were generated.

5. Quality control for raw data

Raw data (raw reads) of fastq format were firstly processed using fastp[2] (v 0.20.0). In this step, clean data (clean reads) were obtained by removing reads containing adapter, reads containing ploy-N and low-quality reads from raw data. At the same time, Q20, Q30 and GC content of the clean data were calculated. All the downstream analyses were based on the clean data.

6. Mapping

Reference genome and gene annotation files were downloaded from genome website directly. Index of the reference genome was built using BWA[3] (v 0.7.12) and clean reads were aligned to the reference genome using BWA mem -k 32 -T 30 -t 4 -M. These reads were then filtered for high quality (MAPQ ≥ 13), Only uniquely mapped reads were used for further analysis.

7. Peak calling

All peak calling was performed with MACS2 [4] (v 2.1.0) using ‘macs2 -q 0.05 -f BAMPE --callsummits --keep-dup all’. By default, peaks with q-value threshold of 0.05 was used for all data sets.

8. Motif analysis

Peaks were adjusted to the same size (500 bp) centered peak summits and motif discoveries of these loci sequences were performed using findMotifsGenome.pl program in HOMER[5] (v 4.9.1) software with ‘-len 8,10,12,14 -gc -size given -homer2 -dumpFasta’.

9. Peak annotation

The position of peak summit around transcript start sites of genes can predict the interaction sites between protein andgene. ChIPseeker[6] (Yu et al., 2015) was used to retrieve the nearest genes around the peak and annotate genomic region of the peak. Peak-related genes can be confirmed by ChIPseeker, and then Gene Ontology (GO) enrichment analysis was performed to identify the function enrichment results. GO enrichment analysis was implemented by the GOseq R package, in which gene length bias was corrected. GO terms with corrected P-value less than 0.05 were considered significantly enriched by peak-related genes[7]. KEGG is a database resource for understanding high-level functions and utilities of the biological system, such as the cell, the organism and the ecosystem, from molecular-level information, especially large-scale molecular datasets generated by genome sequencing and other high-throughput experimental technologies (http://www.genome.jp/kegg/). We used KOBAS software to test the statistical enrichment of peak related genes in KEGG pathways[8].

10. Different peak analysis

Peaks of different groups were merged using ‘bedtools[9] merge’ .We caltulated the mean RPM of each group in the merge peak. Only peaks with fold change of RPM more than 2 were considered as differential peaks. Genes associated with different peaks were identified using ChIPseeker.

References

[1] Kaya-Okur, H. S., S. J. Wu, C. A. Codomo, E. S. Pledger, T. D. Bryson, J. G. Henikoff, K. Ahmad and S. Henikoff (2019). "CUT&Tag for efficient epigenomic profiling of small samples and single cells." Nat Commun 10(1): 1930.

[2] Shifu Chen, Yanqing Zhou, Yaru Chen, Jia Gu; fastp: an ultra-fast all-in-one FASTQ preprocessor, Bioinformatics, Volume 34, Issue 17, 1 September 2018, Pages i884–i890.

[3] Vasimuddin Md, Sanchit Misra, Heng Li, Srinivas Aluru. Efficient Architecture-Aware Acceleration of BWA-MEM for Multicore Systems. IEEE Parallel and Distributed Processing Symposium (IPDPS), 2019.

[4] Yong Zhang et.al. Model-based Analysis of ChIP-Seq (MACS) Genome Biol. 2008; 9(9): R137.

[5] Heinz S, Benner C, Spann N, Bertolino E et al. Simple Combinations of Lineage-Determining Transcription Factors Prime cis-Regulatory Elements Required for Macrophage and B Cell Identities. Mol Cell 2010 May 28;38(4):576-589. PMID: 20513432.

[6] Guangchuang Yu et.al.ChIPseeker: an R/Bioconductor package for ChIP peak annotation, comparison and visualization. Bioinformatics . 2015 Jul 15;31(14):2382-3.

[7] Young, M. D. and M. J. Wakefield, et al. (2010). "Gene ontology analysis for RNA-seq: accounting for selection bias." Genome Biol 11 (2).

[8] Kanehisa, M., M. Araki, et al. (2008). KEGG for linking genomes to life and the environment. Nucleic acids research.(KEGG).

[9] Ramirez, F. and F. Dundar, et al. (2014). "deepTools: a flexible platform for exploring deep-sequencing data." Nucleic Acids Res 42 (Web Server issue): W187-91.